Experiment-9

Write A Python Program To Implement Travelling Salesman Problem

Aim

To write and execute a Python program to solve the Travelling Salesman Problem (TSP) using the brute-force method to find the shortest possible route that visits every city exactly once and returns to the starting city.

Algorithm

1. Start the program.
2. Define the distance matrix representing the distances between cities.
3. Identify the total number of cities.
4. Select the first city as the starting city.
5. Generate all possible permutations of the remaining cities.
6. Add the starting city at the beginning and end of each route.
7. Calculate the total distance of each route.
8. Compare the total distance with the minimum distance found so far.
9. Update the minimum distance if a shorter route is found.
10. Store the route corresponding to the minimum distance.
11. Display the best route and its minimum distance.
12. Stop the program.

Code

ex-9.py - D:/ex-9.py (3.10.0)

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```
from itertools import permutations

# Distance between cities
graph = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

n = len(graph)
cities = list(range(n))

min_cost = float('inf')
best_path = None

# Start from city 0
for path in permutations(cities[1:]):
    route = (0,) + path + (0,)
    cost = 0

    # Calculate total travel cost
    for i in range(len(route) - 1):
        cost += graph[route[i]][route[i + 1]]

    if cost < min_cost:
        min_cost = cost
        best_path = route

# Display result
print("TRAVELLING SALESMAN PROBLEM")
print("---------------------------")
print("Best Route:", end=" ")

for city in best_path:
    print(chr(65 + city), end=" ")

print("\nMinimum Distance:", min_cost)
```

Output

IDLE Shell 3.10.0

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```
Python 3.10.0 (tags/v3.10.0:b494f59, Oct  4 2021, 19:00:18) [MSC v.1929 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: D:/ex-9.py ==========
TRAVELLING SALESMAN PROBLEM
---------------------------
Best Route: A B D C A
Minimum Distance: 80
>>>
```

Result

The Travelling Salesman Problem was successfully implemented using Python 3.10. The program identifies the shortest route as A → B → D → C → A, with a minimum total distance of 80 units.